

Telecom Customer Churn Analysis – Final Report

1. Introduction

Customer churn is a major concern for telecom companies, as acquiring new customers is significantly more expensive than retaining existing ones. Understanding why customers leave and identifying high-risk segments can help organizations design effective retention strategies. This project focuses on analyzing telecom customer churn using SQL and Power BI to uncover meaningful insights and support data-driven decision-making.

2. Objectives of the Study

The main objectives of this project are:

- To analyze customer churn patterns in a telecom dataset
 - To identify key factors influencing customer churn
 - To study churn behavior across demographics, geography, services, and contracts
 - To build an interactive Power BI dashboard for business insights
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3. Dataset Description

The dataset contains telecom customer-level information including:

- Customer demographics (gender, age group, marital status)
- Account information (tenure, contract type, payment method)
- Services subscribed (internet type, phone service, add-on services)
- Churn status and churn category

The data was provided in CSV format and imported into SQL Server for analysis.

4. Tools and Technologies Used

- **SQL Server:** Data cleaning, transformation, and KPI calculation
 - **Power BI:** Data visualization and dashboard creation
 - **Excel / CSV:** Source data handling
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5. Methodology

5.1 Data Cleaning and Preparation

- Removed inconsistencies and handled missing values
- Created derived columns such as tenure groups and age groups
- Standardized categorical values for analysis

5.2 Data Analysis Using SQL

- Calculated key performance indicators (KPIs):
 - Total Customers
 - New Joiners
 - Total Churn
 - Churn Rate
- Analyzed churn across different dimensions such as gender, age group, state, contract type, payment method, tenure group, and services

5.3 Dashboard Development in Power BI

- Designed KPI cards for quick summary insights
- Created interactive visualizations with slicers for filtering by monthly charge range and marital status
- Enabled drill-down analysis for detailed churn exploration

6. Key Performance Indicators (KPIs)

- **Total Customers:** 6.42K
- **New Joiners:** 411
- **Total Churn:** 1.73K
- **Overall Churn Rate:** 27%

7. Analysis and Findings

7.1 Churn by Demographics

- Male and female customers show comparable churn levels
- Older age groups (above 50 years) exhibit a higher churn rate

7.2 Churn by Geography

- Certain states show significantly higher churn rates, indicating region-specific issues

7.3 Churn by Contract Type

- Month-to-month contracts have the highest churn rate (~46.5%)
- One-year and two-year contracts show significantly lower churn

7.4 Churn by Tenure

- Customers with tenure less than 6 months are more likely to churn
- Churn decreases as customer tenure increases

7.5 Churn by Internet and Services

- Fiber optic internet users show higher churn compared to DSL
- Customers without add-on services such as online security or backup tend to churn more

7.6 Churn by Payment Method

- Higher churn observed among customers using mailed checks
- Customers using automated payment methods show lower churn

7.7 Churn Category Analysis

- Competitor-related reasons contribute the most to churn
- Other factors include dissatisfaction, price sensitivity, and attitude

8. Business Insights and Recommendations

- Encourage customers to shift from month-to-month contracts to long-term contracts
- Focus retention efforts on customers in the early tenure stage
- Improve service quality and pricing strategies for fiber optic internet users
- Promote add-on services to enhance customer engagement
- Design region-specific retention strategies for high-churn states

9. Conclusion

This project demonstrates how SQL and Power BI can be effectively used to analyze customer churn and derive actionable business insights. By identifying high-risk customer segments and key churn drivers, telecom companies can proactively implement targeted retention strategies, reduce customer attrition, and improve overall profitability.
